NOVEMBER	TOPIC
6	**DESCRIPTIVE STATISTICS (2/2)** **Relative Position of Data** ~~Percentiles and Quartiles~~ **(RECALL)** **z-scores** **The Empirical Rule and Chebyshev's Theorem,** **The Empirical Rule, Chebyshev's Theorem** ***(неравенствами Чебышёва)***

RELATIVE POSITION OF DATA

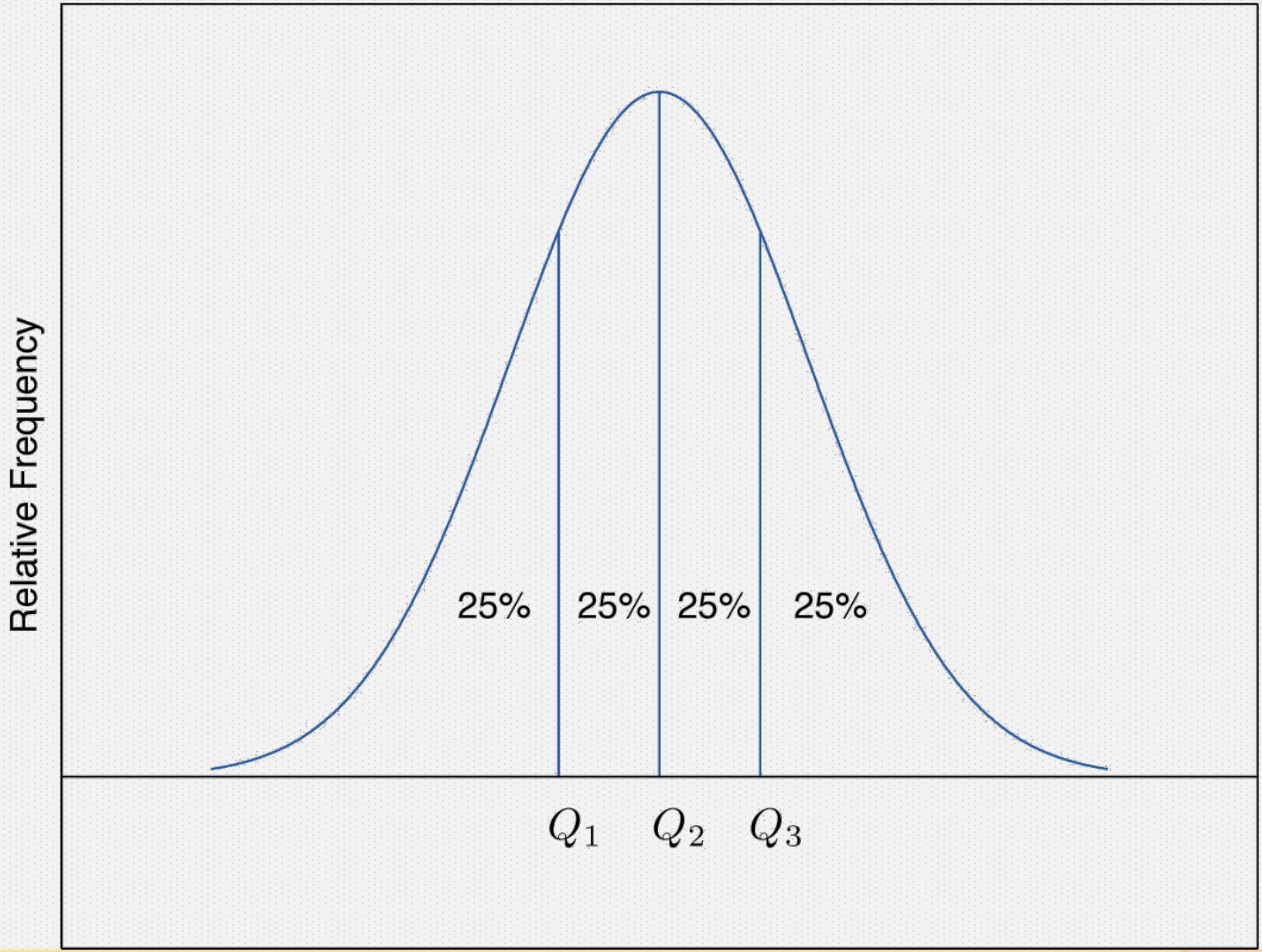
Given an observed value x in a data set, x is the **Pth percentile** of the data if the percentage of the data that are less than or equal to x is P. The number P is the **percentile rank** of x.

The **Pth percentile** cuts the data set in two so that approximately **P%** of the data lie below it and **(100−P)%** of the data lie above it

RELATIVE POSITION OF DATA

The idea of splitting **the data into halves** with **median** or **into quarters** with **quartiles** can be extended to any other fractions. **Deciles** split the data **into tenths**, **percentiles** into **hundredths**. Quantiles are a generalization of this idea to any fraction.

Data Division by Quartiles



For any data set:

1. The second quartile Q2 of the data set is its median.

2. Two subsets:

, i. the lower set: all observations that are , strictly less than Q2;

, ii. the upper set: all observations that are , strictly greater than Q2.

3. The first quartile Q1 of the data set is the median of the lower set.

4. The third quartile Q3 of the data set is the median of the upper set.

RELATIVE POSITION OF DATA

Find the quartiles of the data set of GPAs

3 1.9 2.12 3.88 1.76 2.71 3.33 2.53 1.39 3.71 4.00

Sorting:

1.39 1.76 1.9 2.12 2.53 2.71 3 3.33 3.71 3.88 4.00

Q1 Q2 Q3

RELATIVE POSITION OF DATA

How the quartiles calculated if n is even?

3 1.9 2.12 3.88 1.76 2.71 3.33 2.53 1.39 3.71

Sorting:

1.39 1.76 1.9 2.12 2.53 2.71 3 3.33 3.71 3.88

Q2

RELATIVE POSITION OF DATA

How the quartiles calculated if n is even?

1.39	**1.76**	**1.9**	**2.12**	**2.53**	**2.71**	**3**	**3.33**	**3.71**	**3.88**
1	2	3	4	**5**	6	7	8	9	10

Q2

Q2 = 50 %, n=10

0.5×(n+1) = 5.5

$$[5+0.5] \Rightarrow 2.53 + (2.71 - 2.53) \times 0.5$$

, ,

RELATIVE POSITION OF DATA

How the quartiles calculated if n is even?

1.39	**1.76**	**1.9**	**2.12**	**2.53**	**2.71**	**3**	**3.33**	**3.71**	**3.88**
1	2	3	4	5	6	7	8	9	10

10 %ile

10 %ile, n=10

0.1×(n+1) = 1.1

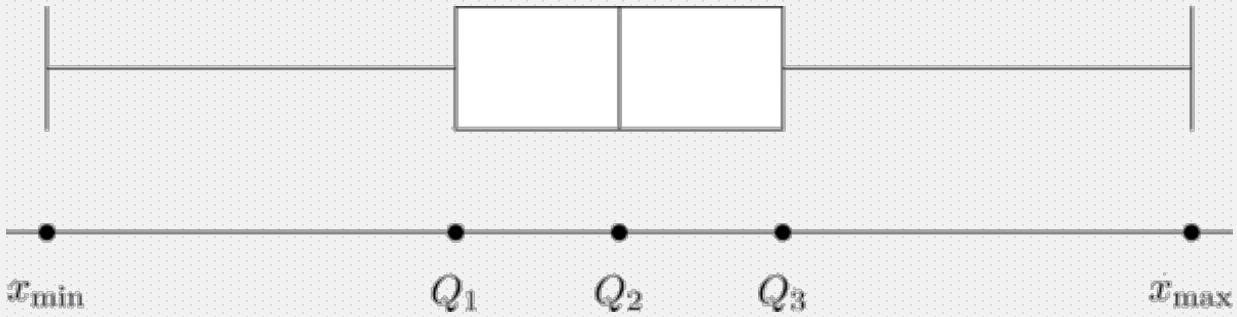
$$[1 + 0.1] \Rightarrow 1.39 + (1.76 - 1.39) \times 0.1$$

10 %ile = 1.427

Box & Whisker Plot

Five-number summary of the data set:

$$\{x_{min}, \mathrm{Q}_1, \mathrm{Q}_2, \mathrm{Q}_3, x_{max}\}$$



Note that the distance from $\boldsymbol{Q_1}$ to $\boldsymbol{Q_3}$ is the length of the interval over which the middle half of the data range. **The interquartile range ($\boldsymbol{IQR}$)** is the quantity:

$$\boldsymbol{IQR = Q_3 - Q_1}$$

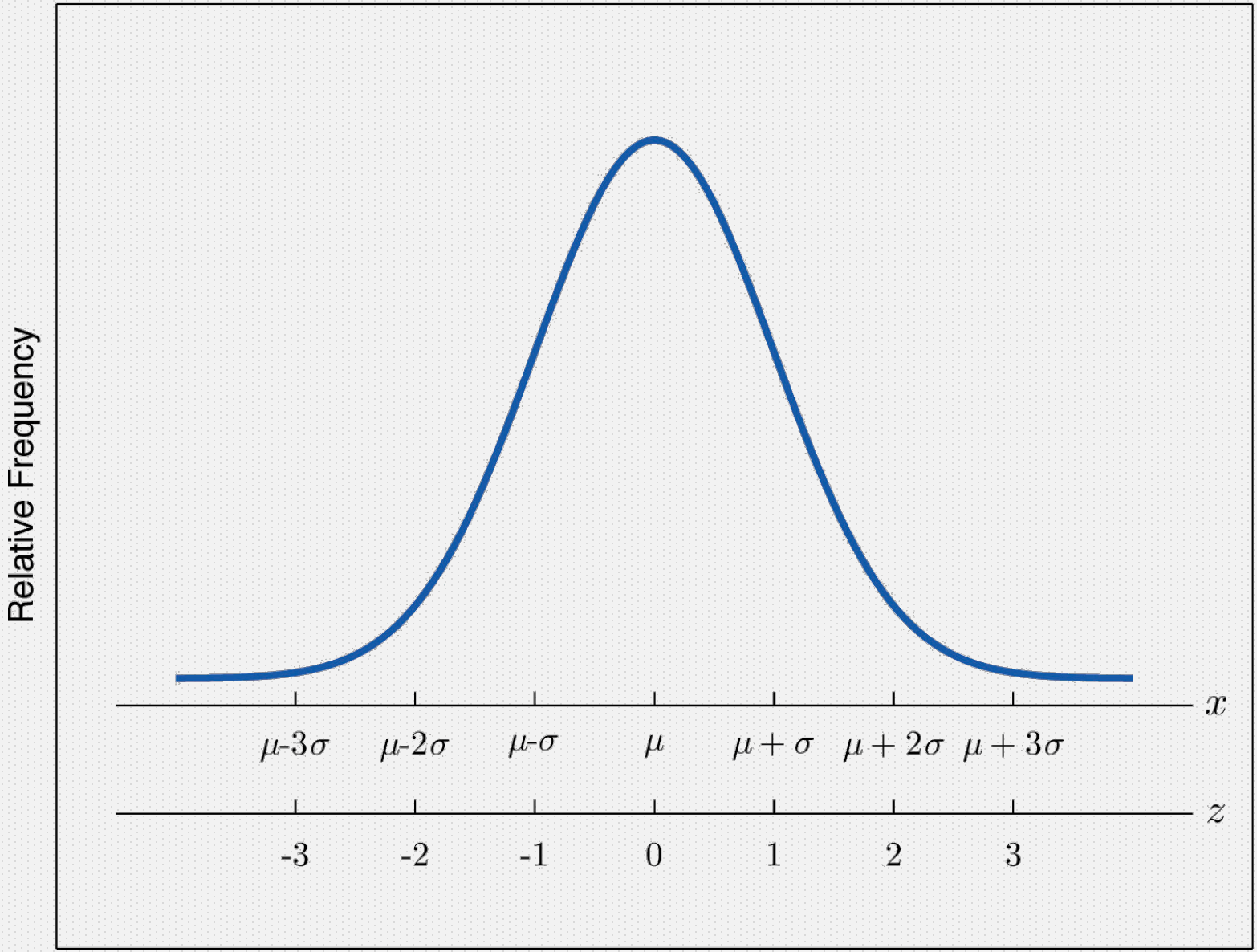
z-scores

Another way to locate a particular observation x in a data set is to compute its distance from the mean in units of standard deviation. Given by:

$$z = \frac{x - \bar{x}}{s} \text{ or } \frac{x - \mu}{\sigma}$$

According to whether data is sample or entire population.

z-scores


Relative Frequency
x
μ-3σ
μ-2σ
μ-σ
μ
$\mu + \sigma$
$\mu + 2\sigma$
$\mu + 3\sigma$
z
-3
-2
-1
0
1
2
3

Example

The following data are sulphur contents (S %) of 30 samples collected from a lignite basin

1.42, 1.64, 2.16, 2.37, 2.49, 2.08, 2.23, 1.53, 1.26, 1.86, 1.25, 1.56, 1.79, 1.98, 1.59, 1.76, 1.67, 1.72, 1.74, 1.92, 1.67, 1.73, 1.60, 1.55, 1.67, 1.89, 2.03, 2.11, 1.90, 1.33

Construct

a) a histogram with 7 intervals starting from 1.20 %
b) a relative histogram with 8 intervals starting from 1.20 %
c) a box plot.

Example

The following data are sulphur contents (S %) of 30 samples collected from a lignite basin

1.42, 1.64, 2.16, 2.37, 2.49, 2.08, 2.23, 1.53, 1.26, 1.86, 1.25, 1.56, 1.79, 1.98, 1.59, 1.76, 1.67, 1.72, 1.74, 1.92, 1.67, 1.73, 1.60, 1.55, 1.67, 1.89, 2.03, 2.11, 1.90, 1.33

Construct

a) a relative histogram with 7 intervals starting from 1.20 % and

b) a box plot.

2.49, 2.37, 2.23, 2.16, 2.11, 2.08, 2.03, 1.98, 1.92, 1.90, 1.89, 1.86, 1.79, 1.76, 1.74, 1.73, 1.72, 1.67, 1.67, 1.67, 1.64, 1.60, 1.59, 1.56, 1.55, 1.53, 1.42, 1.33, 1.26, 1.25

z-scores

Find the z-scores of the data set of GPAs

1.9	**3**	**2.53**	**3.71**	**2.12**	**1.76**	**2.71**	**1.39**	**4**	**3.33**

For 1.9, z-score is calculated as follows:

$\bar{x} = 2.645$
$s = 0.8674$,

$$z = \frac{x - \bar{x}}{s} = \frac{1.9 - 2.645}{0.8674} = \mathbf{-0.8589}$$

1.9	**3**	**2.53**	**3.71**	**2.12**	**1.76**	**2.71**	**1.39**	**4**	**3.33**
-0.86	**0.41**	**-0.13**	**1.23**	**-0.61**	**-1.02**	**0.07**	**-1.45**	**1.56**	**0.79**

KEY TAKEAWAYS

- The percentile rank and z-score of a measurement indicate its relative position with regard to the other measurements in a data set.
- The three quartiles divide a data set into fourths.
- The five-number summary and its associated box plot summarize the location and distribution of the data.

The Empirical Rule

If a data set has an approximately bell-shaped relative frequency histogram, then:

1. Approximately **68%** of the data lie within one standard deviation of the mean, that is, in the interval with endpoints $\bar{x} \pm s$ for samples and with endpoints $\mu \pm \sigma$ for populations;

2. Approximately **95%** of the data lie within two standard deviations of the mean, that is, in the interval with endpoints $\bar{x} \pm 2s$ for samples and with endpoints $\mu \pm 2\sigma$ for populations;

3. Approximately **99.7%** of the data lies within three standard deviations of the mean, that is, in the interval with endpoints $\bar{x} \pm 3s$ for samples and with endpoints $\mu \pm 3\sigma$ for populations;

THE EMPIRICAL RULE AND CHEBYSHEV'S THEOREM

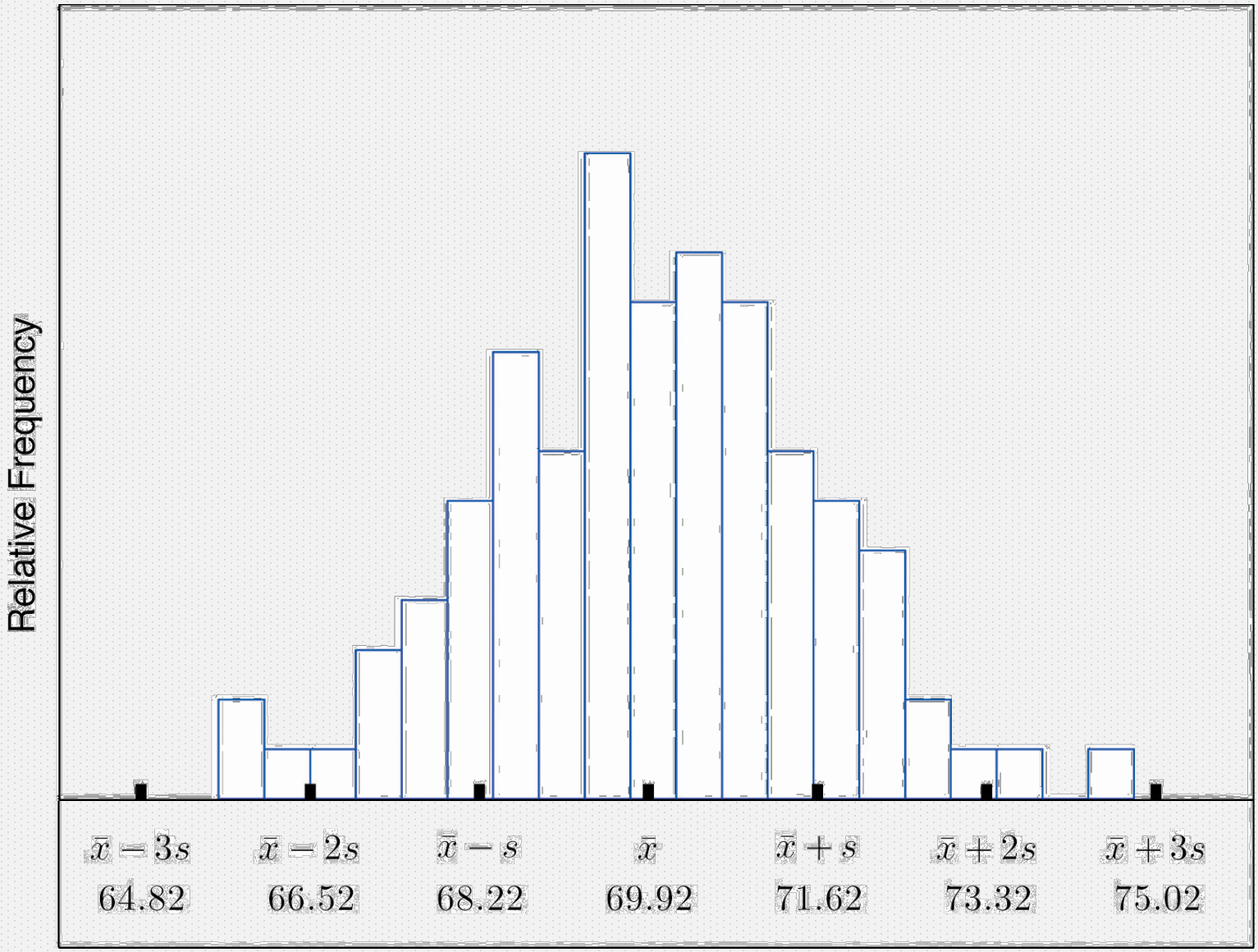
Heights of Men (in)

68.7	72.3	71.3	72.5	70.6	68.2	70.1	68.4	68.6	70.6
73.7	70.5	71.0	70.9	69.3	69.4	69.7	69.1	71.5	68.6
70.9	70.0	70.4	68.9	69.4	69.4	69.2	70.7	70.5	69.9
69.8	69.8	68.6	69.5	71.6	66.2	72.4	70.7	67.7	69.1
68.8	69.3	68.9	74.8	68.0	71.2	68.3	70.2	71.9	70.4
71.9	72.2	70.0	68.7	67.9	71.1	69.0	70.8	67.3	71.8
70.3	68.8	67.2	73.0	70.4	67.8	70.0	69.5	70.1	72.0
72.2	67.6	67.0	70.3	71.2	65.6	68.1	70.8	71.4	70.2
70.1	67.5	71.3	71.5	71.0	69.1	69.5	71.1	66.8	71.8
69.6	72.7	72.8	69.6	65.9	68.0	69.7	68.7	69.8	69.7

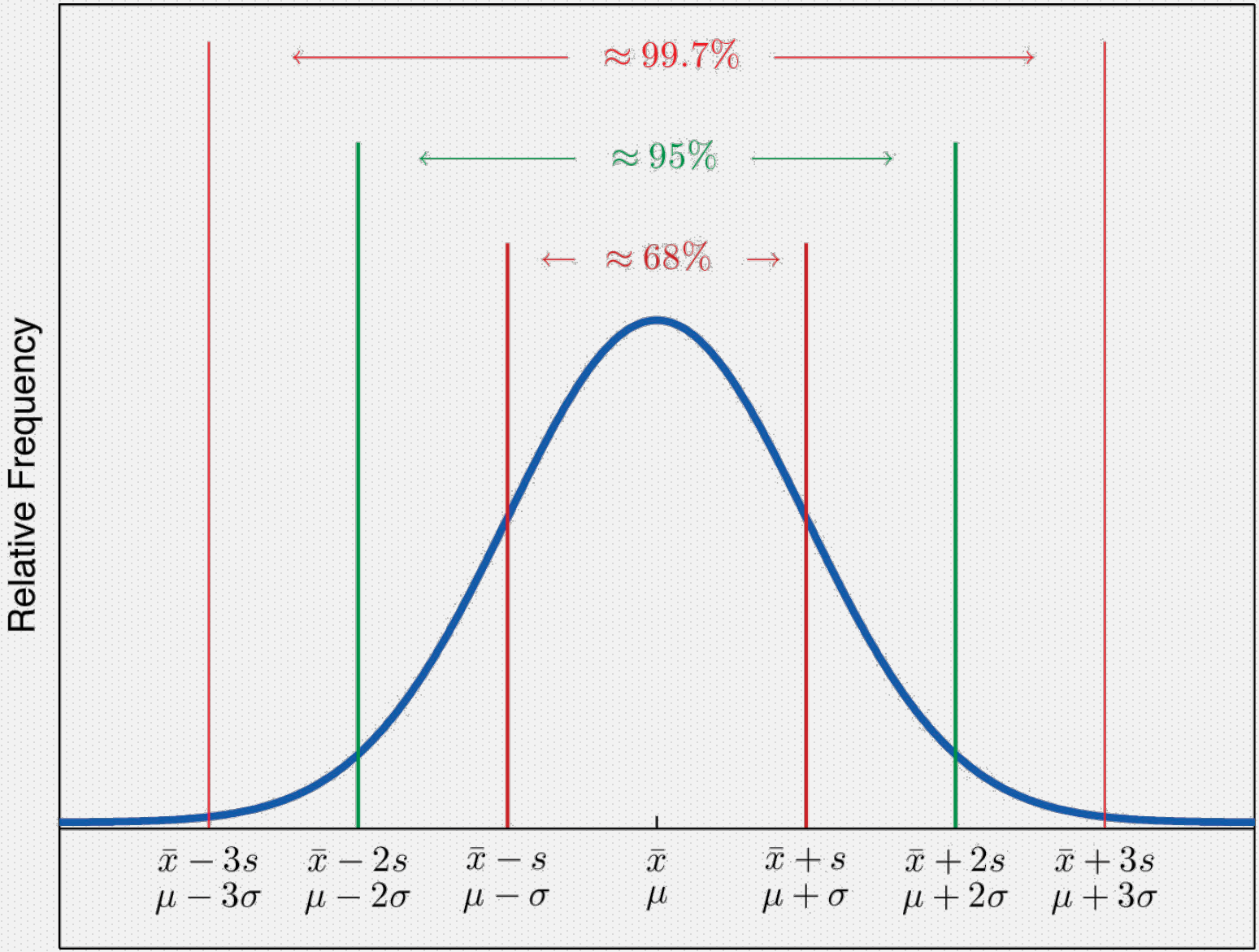
$$\bar{x} = 69.92$$

$$s = 1.70$$

THE EMPIRICAL RULE AND CHEBYSHEV'S THEOREM



THE EMPIRICAL RULE AND CHEBYSHEV'S THEOREM

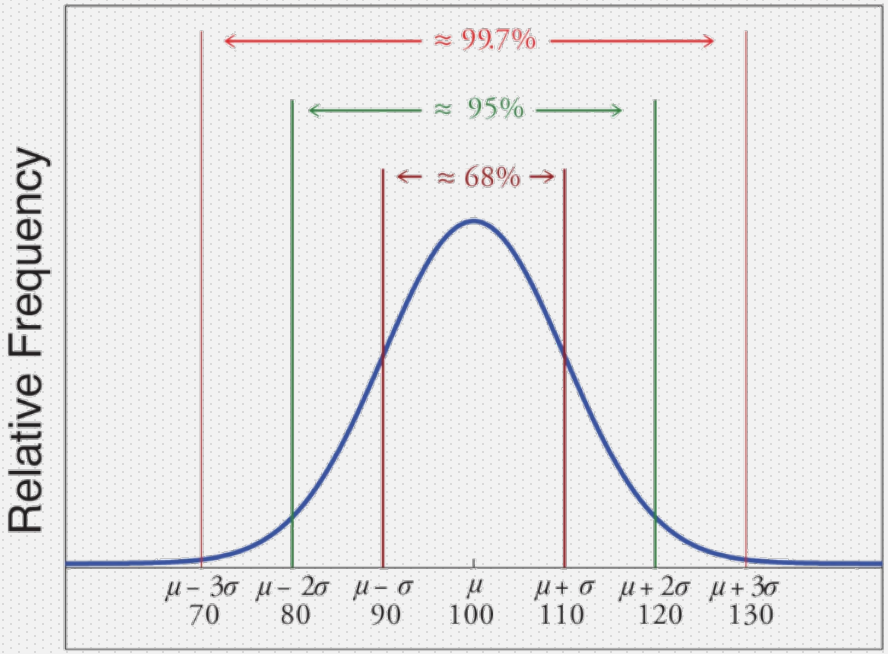


The Empirical Rule

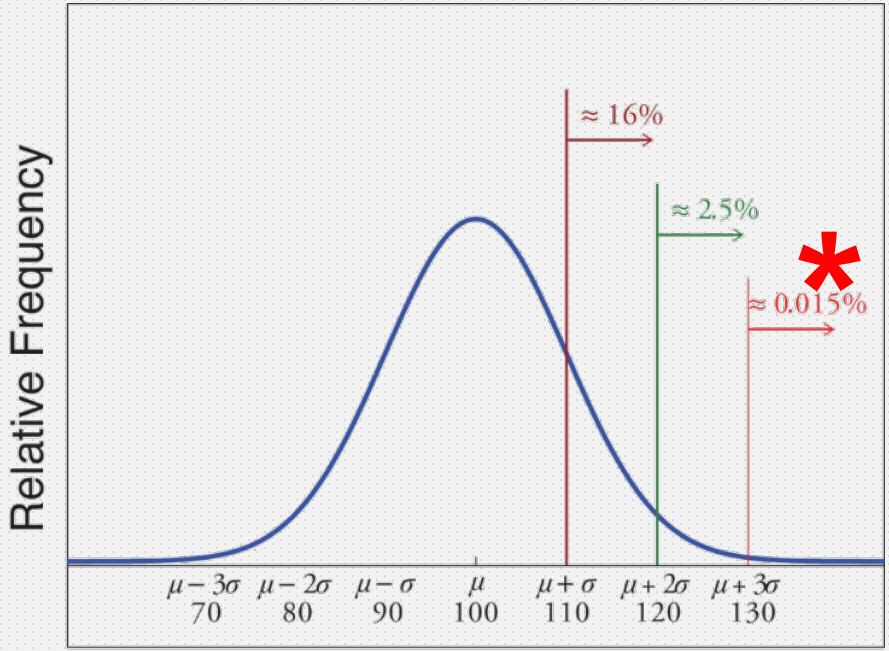
- Data distribution must be approximately *bell-shaped*
- The percentages are only *approximately* true.

The Empirical Rule does not apply to data sets with severely asymmetric distributions.

The Empirical Rule



(a) Whole Spectrum



(b) Higher End

*** 0.15%**

The Chebyshev's Rule
неравенствами Чебышёва

For any numerical data set,

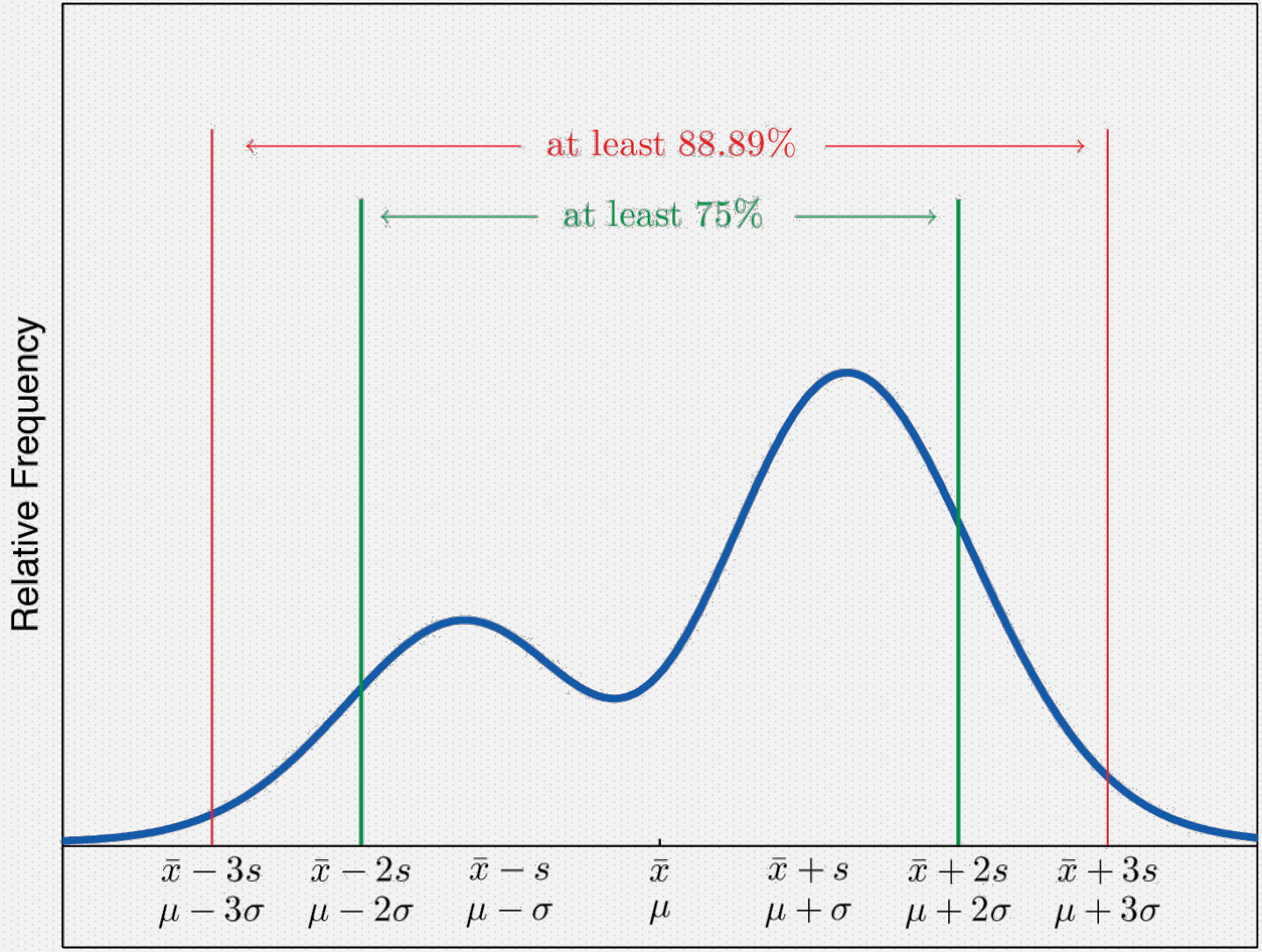
1. at least **75%** of the data lie within two standard deviations of the mean, that is, in the interval with endpoints $\bar{x} \pm 2s$ for samples and with endpoints $\mu \pm 2\sigma$ for populations
2. at least **88.89%** of the data lie within three standard deviations of the mean, that is, in the interval with endpoints $\bar{x} \pm 3s$ for samples and with endpoints $\mu \pm 3\sigma$ for populations

The Chebyshev's Rule неравенствами Чебышёва

For any numerical data set,

3. at least $1 - \frac{1}{k^2}$ of the data lie within k standard deviations of the mean, that is, in the interval with endpoints $\bar{x} \pm ks$ for samples and with endpoints $\mu \pm k\sigma$ for populations, where k is any positive whole number that is greater than 1.

The Chebyshev's Rule
неравенствами Чебышёва



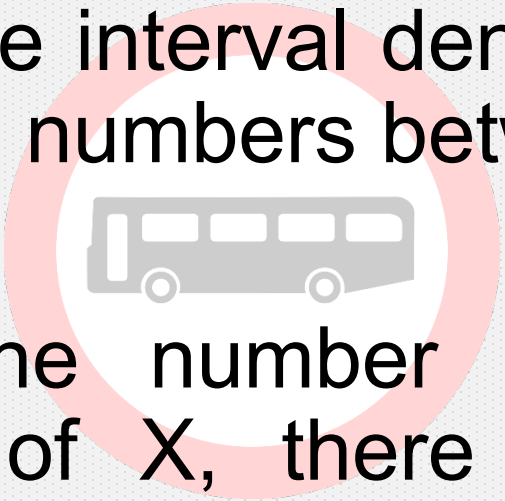
KEY TAKEAWAYS

- **The Empirical Rule** is an approximation that applies only to data sets with a bell-shaped relative frequency histogram. It estimates the proportion of the measurements that lie within **one, two, and three standard deviations** of the mean.
- **Chebyshev's Theorem** is a fact that applies to all possible data sets. It describes the minimum proportion of the measurements that lie must within **one**, **two**, **or more standard deviations** of the mean.

NOVEMBER	TOPIC
6	CONTINUOUS RANDOM VARIABLES

CONTINUOUS RANDOM VARIABLES

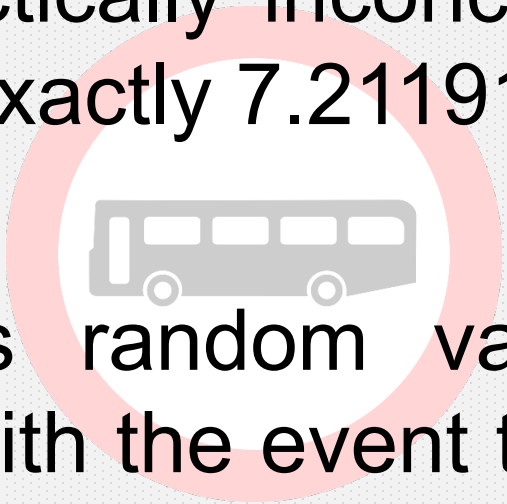
Suppose X denotes the length of time a commuter just arriving at a bus stop has to wait for the next bus. If buses run every 30 minutes without fail, then the set of possible values of X is the interval denoted [0,30], the set of all decimal numbers between 0 and 30.



But although the number 7.211916 is a possible value of X, there is little or no meaning to the concept of the probability that the commuter will wait precisely 7.211916 minutes for the next bus.

CONTINUOUS RANDOM VARIABLES

If anything the probability should be zero, since if we could meaningfully measure the waiting time to the nearest millionth of a minute it is practically inconceivable that we would ever get exactly 7.211916 minutes.



With continuous random variables one is concerned not with the event that the variable assumes a single particular value, but with the event that the random variable assumes a value in a particular interval.

CONTINUOUS RANDOM VARIABLES

What is the probability that the commuter's waiting time is less than 10 minutes, or is between 5 and 10 minutes?

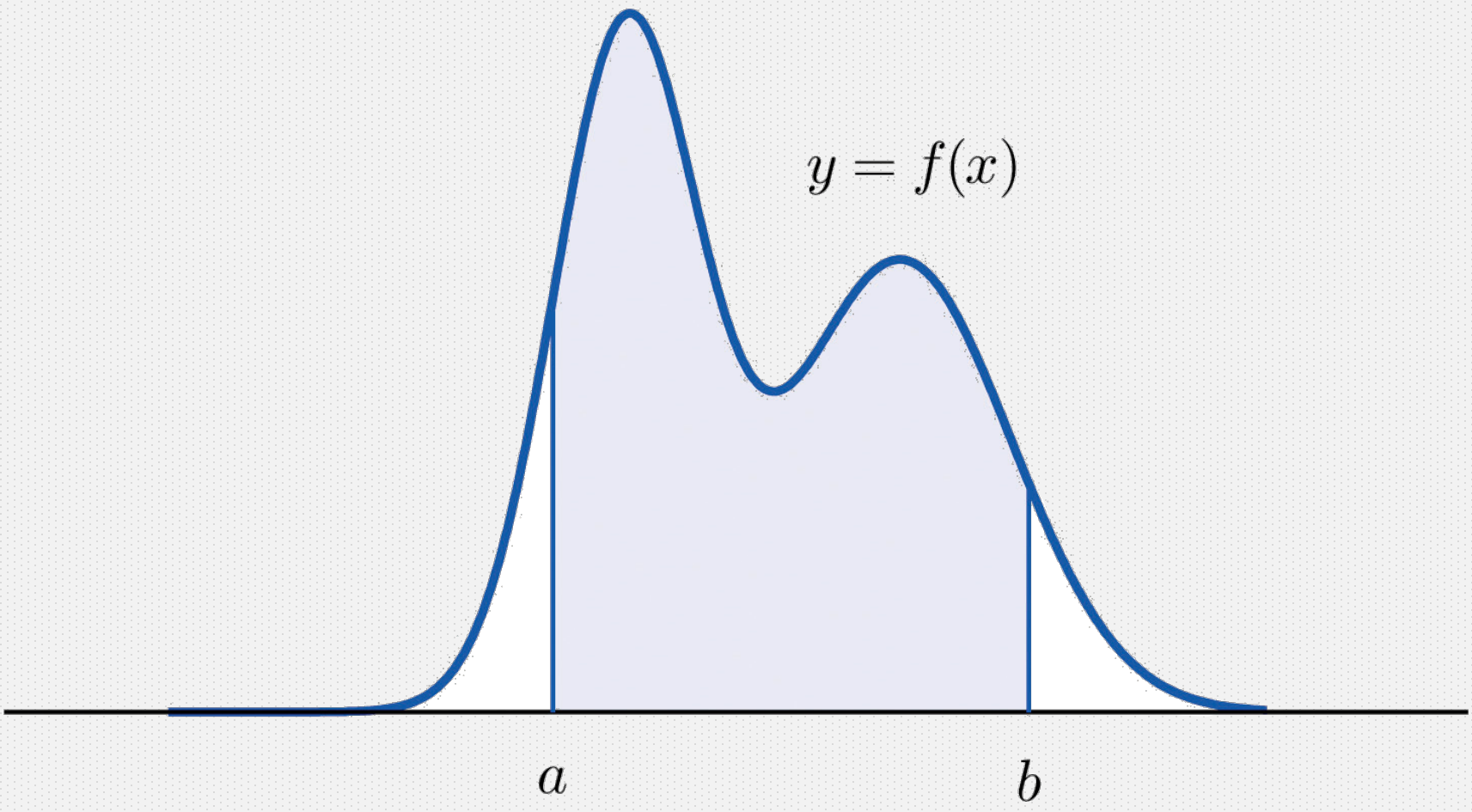
CONTINUOUS RANDOM VARIABLES

The probability distribution of a continuous random variable X is an assignment of probabilities to intervals of decimal numbers using a function f(x), called a **density function**, in the following way:

The probability that X assumes a value in the interval [a,b] is equal to the area of the region that is bounded above by the graph of the equation y = f(x), bounded below by the x-axis, and bounded on the left and right by the vertical lines through a and b.

CONTINUOUS RANDOM VARIABLES

$$P(a < X < b) = \text{area of shaded region}$$



NORMAL DISTRIBUTIONS

"Bell curve": It is the graph of a specific density function f(x) that describes the behavior of continuous random variables. The formula for f(x) contains two parameters μ and σ that can be assigned any specific numerical values.

$$\mathrm{f(x)} = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\mu - x)^2 / \sigma^2}$$

where, $\pi \approx 3.14159$ and $e \approx 2.71828$